



# Multi-scale feature pyramid fusion network for medical image segmentation

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Received: 2 March 2022 / Accepted: 11 August 2022 / Published online: 30 August 2022  
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## Abstract

**Purpose** Medical image segmentation is the most widely used technique in diagnostic and clinical research. However, accurate segmentation of target organs from blurred border regions and low-contrast adjacent organs in Computed tomography (CT) imaging is crucial for clinical diagnosis and treatment.

**Methods** In this article, we propose a Multi-Scale Feature Pyramid Fusion Network (MS-Net) based on the codec structure formed by the combination of Multi-Scale Attention Module (MSAM) and Stacked Feature Pyramid Module (SFPM). Among them, MSAM is used to skip connections, which aims to extract different levels of context details by dynamically adjusting the receptive fields under different network depths; the SFPM including multi-scale strategies and multi-layer Feature Perception Module (FPM) is nested in the network at the deepest point, which aims to better focus the network's attention on the target organ by adaptively increasing the weight of the features of interest.

**Results** Experiments demonstrate that the proposed MS-Net significantly improved the Dice score from 91.74% to 94.54% on CHAOS, from 97.59% to 98.59% on Lung, and from 82.55% to 86.06% on ISIC 2018, compared with U-Net. Additionally, comparisons with other six state-of-the-art codec structures also show the presented network has great advantages on evaluation indicators such as Miou, Dice, ACC and AUC.

**Conclusion** The experimental results show that both the MSAM and SFPM techniques proposed in this paper can assist the network to improve the segmentation effect, so that the proposed MS-Net method achieves better results in the CHAOS, Lung and ISIC 2018 segmentation tasks.

**Keywords** Medical image segmentation · Multi-Scale attention module · Stacked feature pyramid module · Feature perception module

## Introduction

In the field of medical image processing, accurate segmentation of high-risk abdominal organs [1, 2], skin lesions [3, 4] and other pathologies [5–7] is extremely important for medical image analysis. Recently, 3D and 4D have been reported for detection and segmentation tasks, such as lung cancer

and brain lesions [8, 9], which demonstrate the advantages of extracting rich 3D contextual information from volumetric images, but its reliance on computational resources is obvious [10]. How CNN networks scale up for 3D or 4D tasks has become the new technology trend [11, 12]. Lei Mou et al. extend the CS-Net to CS<sup>2</sup>-Net for 3D vessel segmentation task, by replacing normal convolution with high-dimensional convolution [13]. For 3D Xenopus kidney segmentation, Ozgün Cicek et al. [14] propose a 3DU-Net structure by extending the u-net network previously proposed by Ronneberger. Furthermore, Sil C. Van De Leemput et al. converted the 4D dataset into 3D sequence data and fed it into a 3D network for multi-class brain tissue segmentation [15]. Andriy Myronenko et al. redesigned the encoder-decoder network with 4D convolution kernels for accurate segmentation of the ventricle from CT [16]. It can be seen that a

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common approach to constructing CNN networks for 3D or 4D medical image segmentation is to try high-dimensional convolutional operations on top of 2D networks [17, 18]. Therefore, researches on 2D CNNs are very active in medical segmentation because of their low computational cost and scalability for higher dimensional tasks [10]. Accurate segmentation of target organs from CT images using 2D convolutions is an important step to ensure accurate treatment [19]. However, accurate segmentation of target regions is a very challenging task due to blurring and low contrast between human organs (lungs, adrenal glands, and spleen) [20]. Over the past few decades, many approaches to disease segmentation have been reported, which can be reviewed comprehensively in [21, 22].

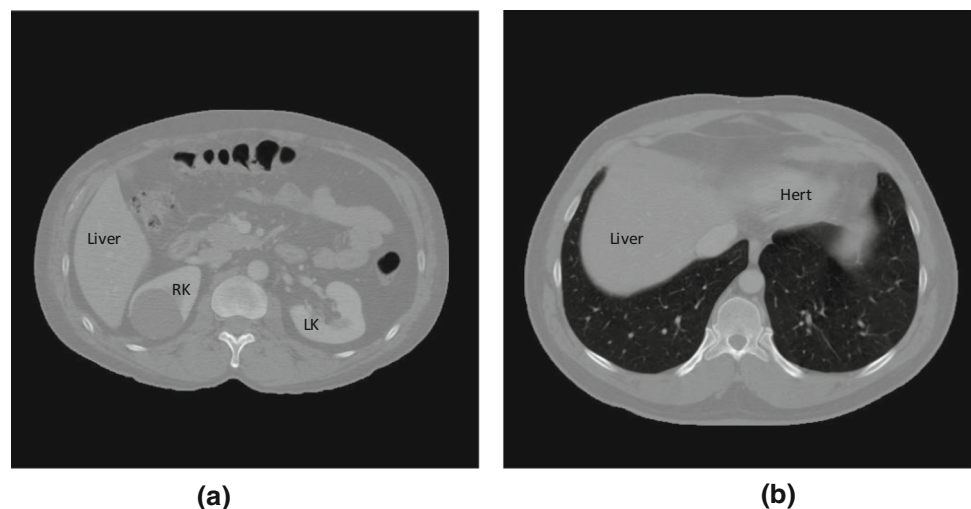
In recent years, with the rise of deep learning, convolutional neural networks have developed rapidly in the field of computer vision [23, 24], especially in medical image segmentation [25, 26]. Ronneberger et al. (2015) [27] proposed a U-shaped structure network, which mainly introduced skip connections between the encoder and decoder to preserve the extracted details, thereby improving the segmentation performance of the network. (2018) [28] proposed U-Net++, which changed the skip connection of U-Net [27] from the original direct connection mode to the dense connection mode. The network can automatically extract different features during training to optimize segmentation efficiency. Chen et al. (2018) [29] proposed the ResUnet model, which utilizes the residual mechanism [30, 31] to improve the feature extraction rate and alleviate the gradient explosion problem during segmentation. Later, many attention mechanisms [32] were proposed to assign weights to features at different levels to improve segmentation of complex regions. Octet et al. (2018) [33] proposed an Attention-Unet structure to improve segmentation accuracy by increasing the sensitivity of pixels in the region of interest. Chen Ye and others.

(2019) proposed [34] Channel-Unet to process spatial channel convolution information, which can effectively suppress over-segmentation and under-segmentation in the segmentation process. Meanwhile, Gu et al. (2019) [35] designed a CE-Net network model to alleviate the problem of information loss during feature extraction by adjusting the receptive field region through an expanded convolution mechanism. Mueller et al. (2019) [36] proposed CS-Net to effectively combine global and local features by extracting spatial and channel information. To explore higher liver segmentation performance, A. Emre Kavur et al. studied the ensemble strategy of six independent deep learning models by using two fusion algorithms, namely: Majority Voting (MV) and Simultaneous Truth and Performance Level Estimation (STAPLE). It is found that the ensemble-based fused deep learning models outperform any single network in segmentation tasks [37, 38].

Although these methods greatly improve the segmentation effect, there are still huge challenges in blurring the edges (see Fig. 1a) and organs with low contrast between adjacent organs (see Fig. 1b). Firstly, most networks cannot effectively focus on target pixels when dealing with low-contrast targets. Incorrect extraction of feature information leads to over-segmentation problems. Secondly, most networks usually use fixed receptive fields to extract features. When extracting the feature information of the fuzzy area, the network cannot dynamically adjust the receptive field to adapt to different detailed features.

This paper proposes a new Multi-Scale Feature Pyramid Fusion Network (MS-Net), which is mainly used to segment some organs with low contrast and fuzzy edges. MS-Net mainly includes two new modules, namely Multi-Scale Attention Module (MSAM) and Stacked Feature Pyramid Module (SFPM). Firstly, in order to better extract high-level semantic information of fuzzy edge regions, the proposed

**Fig. 1** The main factors affecting segmentation: **a** blurred edges, **b** low contrast with other organs



MSAM can effectively adjust the receptive field and obtain feature information of different scales. Secondly, the SFPM is composed of three parallel filters with different receptive fields and three Feature Perception Modules (FPM). The SFPM can better learn the feature information of the region of interest, so that the network can better pay attention to the target pixel. Finally, the proposed MS-Net was evaluated on three challenging tasks: liver segmentation in the CHAOS dataset [39], lung segmentation in the Lung dataset [40], and skin lesion segmentation in the ISIC 2018 dataset [41, 42]. The main work of this paper includes the following three aspects.

- (1) This paper proposes a new MS-Net to solve the challenge of segmentation of medical lesions. The network can accurately segment target areas in complex and fuzzy areas.
- (2) At the skip connection of the network, a new MSAM mechanism is designed to Adjust the receptive field under different network depths. Extract multi-scale and contextual information more effectively.
- (3) The SFPM is designed to enhance the sensitivity of the network to target pixels, expand the width and depth of the network, and enhance the adaptability of the network to different scales.

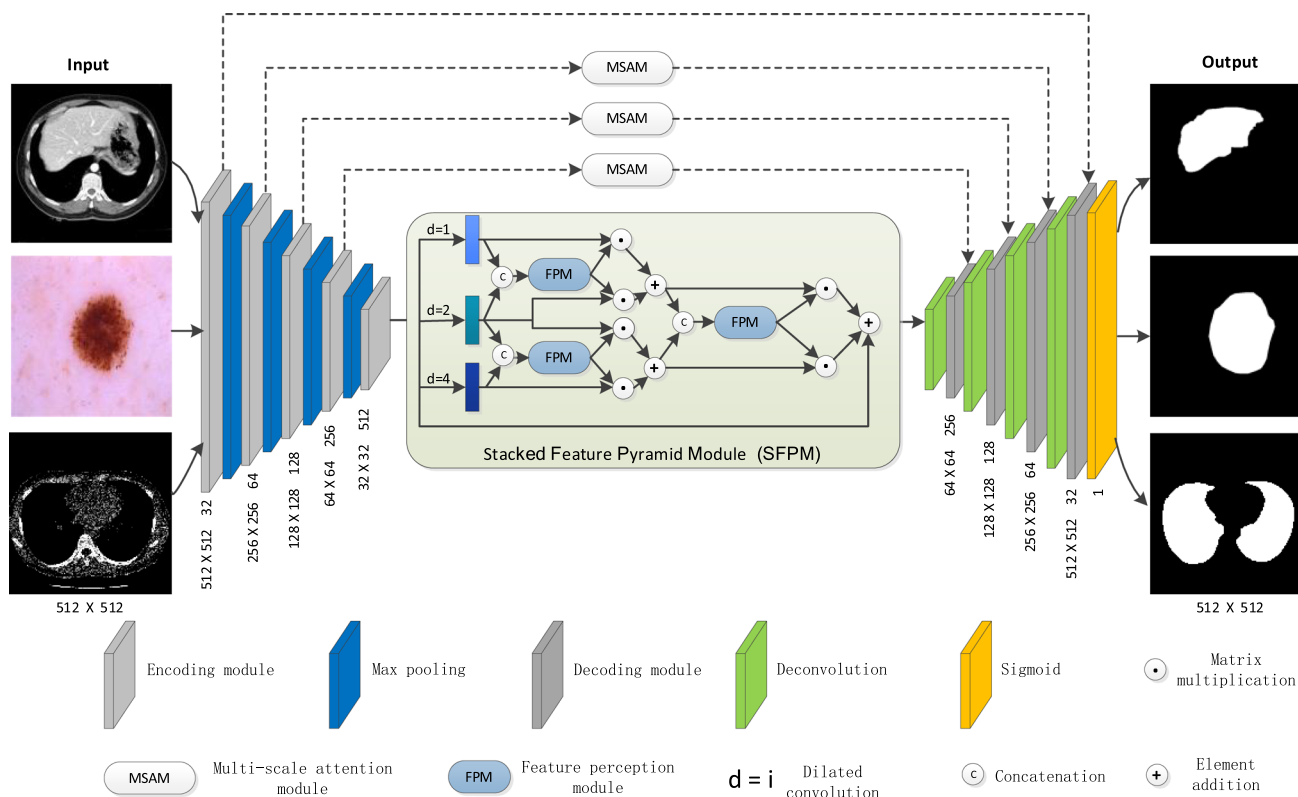
The rest of this article is organized as follows. In the second part, we introduced our MS-Net in detail. The third part describes the dataset and evaluation indicators. In the fourth part, we compare and analyze the experimental results. Finally, Section five summarizes the article.

## Method

In this part, we will introduce MS-Net based on the codec structure and its main components, namely MSAM for acquiring context information, SFPM for feature extraction and FPM for adaptive learning weights.

### Network architecture

The network structure proposed in this paper is shown in Fig. 2. Our MS-Net backbone network is based on the encoding and decoding structure of U-Net [27], which has been widely used in medical segmentation tasks. The encoding module is composed of continuous convolution and maximum pooling, and is mainly used to extract feature pixels of interest. The decoding module is composed of a series of convolution and deconvolution, and gradually maps the extracted feature map to the size of the original image.



**Fig. 2** MS-Net structure. There are three modules in MS-Net

MS-Net mainly includes two new modules, namely MSAM and SFPM. The MSAM is composed of dilated convolution strategies with different expansion rates and a residual mechanism, which aims to improve the sensitivity of feature pixels in the blurred edge area. Since the receptive field varies with network depths, a dynamically adjustable receptive field can be beneficial to segmentation performance enhancement. In this regard, MSAM adopts a method of extracting features in parallel by expanding the convolution of two different receptive fields. Additionally, layers of different depth are assigned with corresponding receptive fields for better adaptability.

They are the encoding and decoding structure, the MSAM for skip connection, and the SFPM for feature depth extraction.

The SFPM combines multi-scale strategies and FPM to increase the weight of target features. This module is mainly used to enhance the feature information of the target organ in the features extracted by the coding layer. Then, the sub-module FPM is used to adaptively train the weights from the features extracted from the adjacent receptive fields. Through SFPM, the network may be able to make the target organ more prominent in some adjacent organs with lower contrast.

### Multi-scale attention module

When the encoder extracts contextual information from the input image, as the network deepens, it becomes less sensitive to detailed information. Skipping connections is a common way to alleviate this problem, but it also generates irrelevant noise [54]. In this paper, we propose a multi-scale strategy based MSAM that uses dilated convolutions to efficiently extract different levels of detail features, as shown in Fig. 3.

MSAM is designed based on the channel attention mechanism. The module first pools the feature map to reduce the dimensionality and remove redundant information, and filters the information through convolution. Then, two dilated convolutions of different receptive fields are used in parallel to extract different levels of multi-scale information. It should be noted that here we use dilated convolution combinations with different expansion rates to adapt to different network depths ( $d = 1, d = 3$  for the second-stage of the codec structure;  $d = 3, d = 5$  for the third-stage; and  $d = 5, d = 7$  for the fourth stage). The MASM of the second stage is shown in Fig. 3. Finally, after convolution and fusion of the extracted multi-scale feature information of different levels, the initial features are fused in the form of residual connection. The whole module can be expressed as Eq. (1).

$$F_{out} = ReLU \left( Multi \left( Conv2 \left( C \left( \begin{matrix} d=1 \\ F \\ Conv1 \end{matrix} ; \begin{matrix} d=3 \\ F \\ Conv1 \end{matrix} \right) \right), F \right) \right) \quad (1)$$

Among them,  $F$  represents the input feature map;  $Conv1$  represents two cycles of  $3 \times 3$  regular convolution and ReLU processing;  $d$  is the dilated convolution processing;  $C$  represents the cascade operation;  $Conv2$  indicates operation consisting of  $3 \times 3$  regular convolution, BN, and ReLU. Multi indicates the multiplication operation.

We introduce MSAM as a skip connection into the deeper three layers of the network.

### Stacked feature pyramid module

For effectively alleviate the problem of low contrast between the target and other regions in the segmentation task, we adopt a parallel expansion convolution strategy to extract and integrate different detailed features at the same time. Therefore, the SFPM is an attention module designed based on a multi-scale strategy and FPM, which aims to identify target areas in certain low-contrast environments as shown in Fig. 4.

The realization of SFPM mainly includes the following steps. Firstly, the module uses dilated convolutions with dilation ratios of 1, 2 and 4 in parallel to process the information extracted from the coding layer, and uses adjacent receptive fields to enhance different detailed features. Secondly, the FPM is used to enhance the intimacy between pixels to make the target area more prominent. The statement is as follows:

$$(f_1 f_2) = FPM(C(F_1; F_2)) \quad (2)$$

$F_1$  and  $F_2$  indicate that the feature map  $F$  has undergone dilated convolution processing with dilation rates of 1 and 2, respectively.  $C$  is the cascade operation. FPM is used for adaptive training weights.  $f_1$  and  $f_2$  represent the weights assigned to the feature map after calling the FPM to learn. Then, the learned weights with adjacent feature maps are fused in the form of weighted summation.

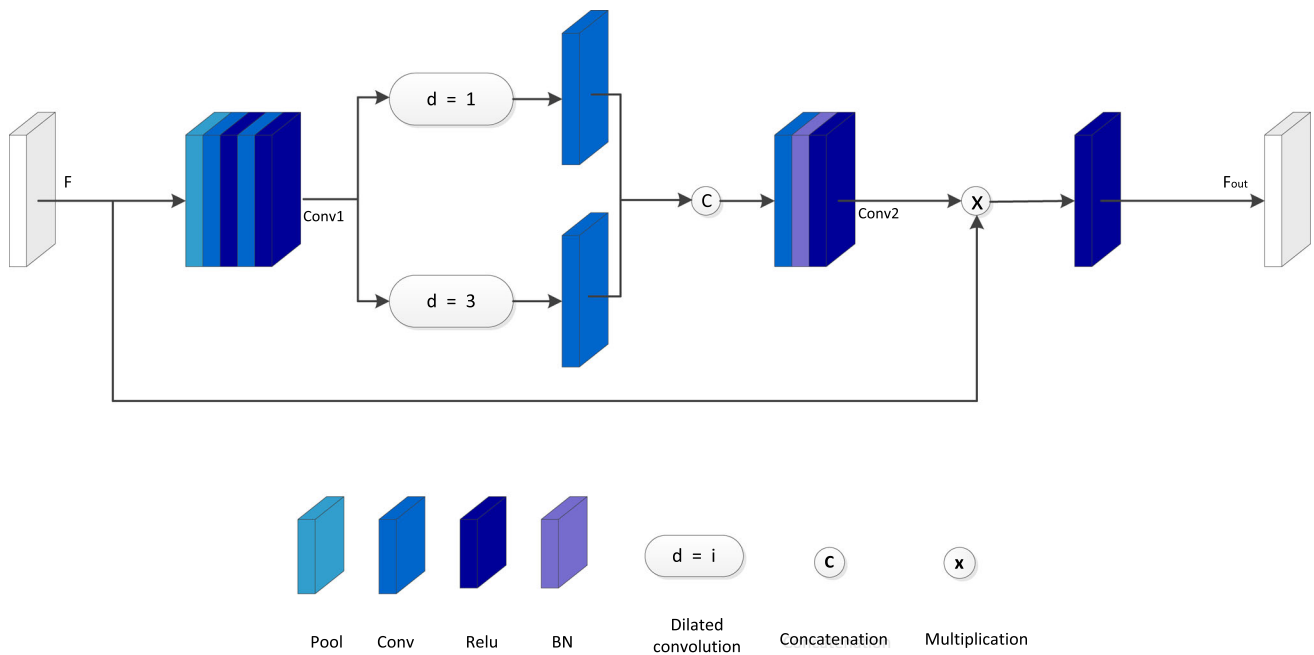
$$F_{12} = F_1 \odot f_1 + F_2 \odot f_2 \quad (3)$$

$$F_{24} = F_2 \odot f_2 + F_4 \odot f_4 \quad (4)$$

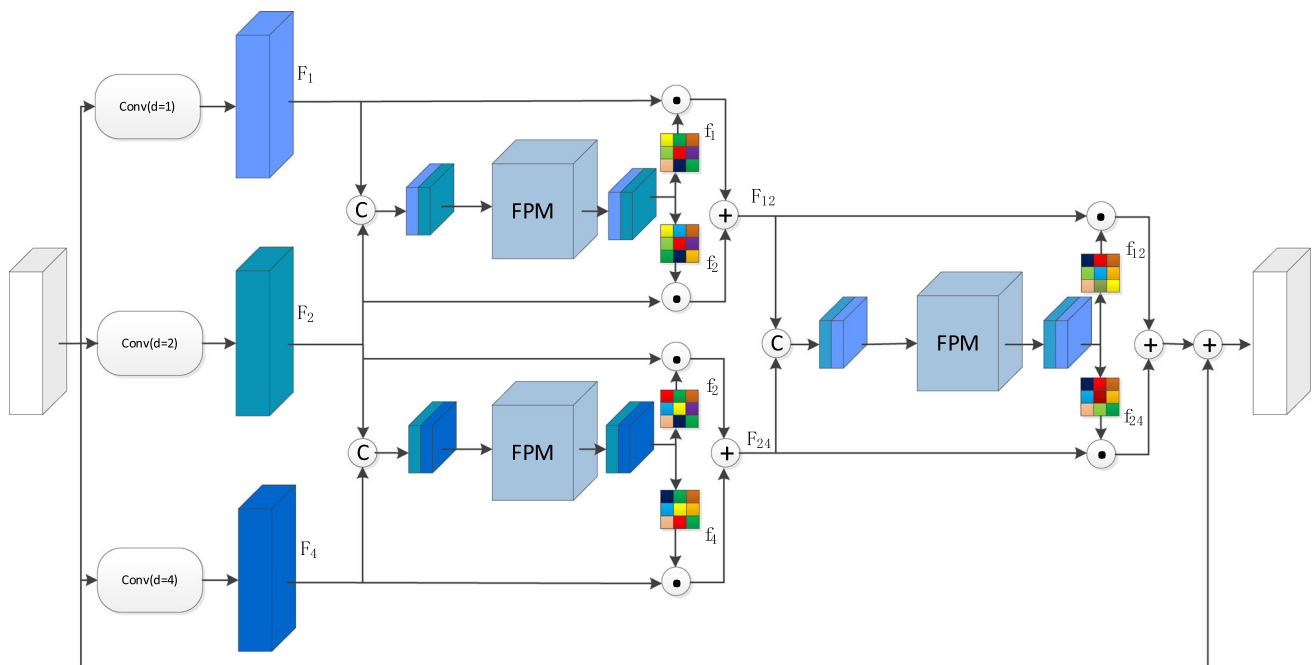
In the above formula,  $\odot$  represents matrix multiplication. After the same operation, we can get the coefficient  $F_{24}$ . After the above processing, two features ( $F_{12}$  and  $F_{24}$ ) are obtained through the fusion processing of adjacent receptive fields. Finally, we use FPM again to process  $F_{12}$  and  $F_{24}$  to extract deeper details.

$$(f_{12}, f_{24}) = FPM(C(F_{12} F_{24})) \quad (5)$$

$$F_{out} = F + (F_{12} \odot f_{12} + F_{24} \odot f_{24}) \quad (6)$$



**Fig. 3** Multi-Scale Attention Module



**Fig. 4** Stacked feature pyramid module, composed of three parallel dilated convolutions of different scales and three FPMs

where  $f_{12}$  and  $f_{24}$  are the learning weights of features  $F_{12}$  and  $F_{24}$ , respectively.

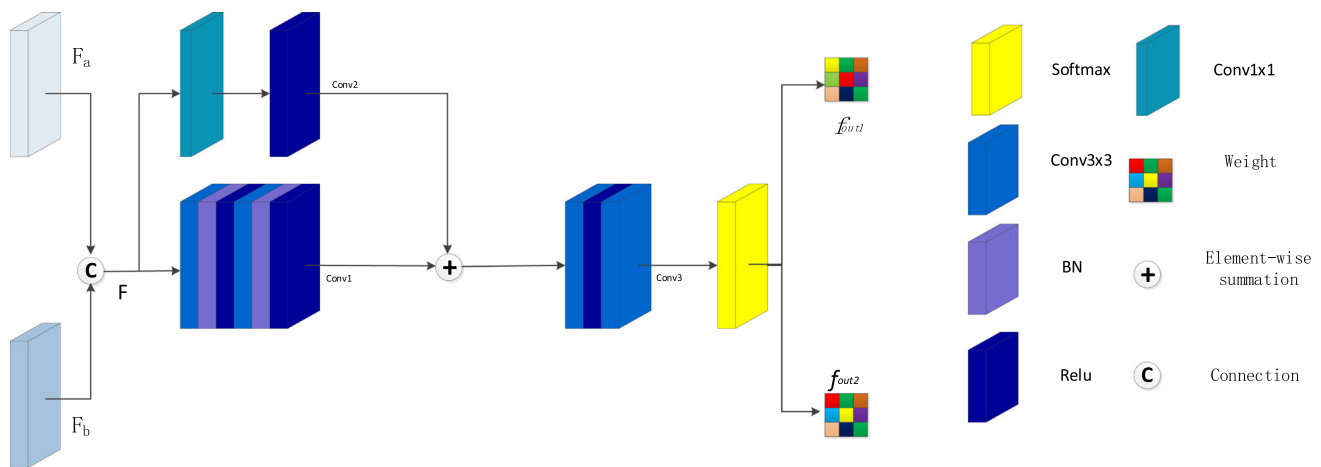
### Feature perception module

In the SFPM, we use a sub-module called FPM, which improves the ability of the network to capture features of interest. The FPM is based on the channel attention mech-

anism, dynamically selects interesting features of different scales, and can adaptively increase the weight of features of interest.

First, The FPM aggregates two feature maps from different receptive fields, as shown in Fig. 5. It should be noted that the residual connection mechanism is introduced in the FPM to prevent the gradient from disappearing. Then, regular convolution is used to extract high-level semantic feature





**Fig. 5** Schematic diagram of Feature Perception Module

information.  $1 \times 1$  convolution is used at the residual connection to reduce network parameters and prevent network overfitting. Finally, the extracted information is convolved, and then sent to the Softmax layer to obtain two trained weights. The FPM can be summarized as formula (7) (8):

$$F = C(F_a F_b) \quad (7)$$

$$(f_{out1} + f_{out2}) = S(Conv3(F_{Conv1} + F_{Conv2})) \quad (8)$$

$F_a$  and  $F_b$ , respectively, represent feature maps of different scales, and  $C$  represents cascading operations. *Conv1* block represents twice operations consisting of  $3 \times 3$  regular convolution, BN and ReLU. *Conv2* block represents  $1 \times 1$  convolution and ReLU sequential operation. *Conv3* block represents  $3 \times 3$  convolution, ReLU and  $3 \times 3$  convolution multiplication operation, which purpose is to extract features.  $S$  represents the Softmax layer, and  $f_{out1}$  and  $f_{out2}$  are the weights after learning  $F_a$  and  $F_b$ , respectively.

## Experimental scheme

### Datasets

In order to train, test and evaluate medical segmentation algorithms, high-quality ground truth is necessary to establish a medical dataset, which are usually annotated by human experts according to relevant rules [43]. The inaccuracy or ineffectiveness of medical ground truth data also limits the credibility of the assessment [44]. To obtain ground truth with high accuracy, independent annotations must be carried out by multiple medical experts and further cross-validated through statistical methods [44, 45]. Furthermore, numerous researchers have developed software platforms to assist, standardize, and accelerate the aforementioned annotation

process [45–47]. However, the acquisition and reliable annotation of medical datasets remains a challenging task that requires a great deal of work and experience. Fortunately, many researchers have contributed to establish many public datasets, thus three popular datasets are adopted to evaluate the proposed network, including: the CHAOS [39], the Kaggle Lung [40] and the ISIC 2018 Abdominal Organs [41, 42]. In our experiment, we uniformly adjusted the resolution of the three datasets to  $512 \times 512$ , and the image format was set to jpg format.

**CHAOS** The CHAOS dataset collected at the Department of Radiology, Dokuzulu University Hospital, Izmir, Turkey, followed by an open challenge dataset in 2018. Generally, in open segmentation challenges, there is often a form called "peeking effect" to get higher scores [49]. Peeking effect refers to iteratively report results by tuning model parameters, which is considered to be an injustice to the majority of participants in a challenge. Therefore, organizers of the CHAOS Challenge have taken various strategies to avoid the effect, such as: limiting the number of submissions, requiring the institutional information, and attaching a technical documentation of the method [50]. In the newly reported work on abdominal organ segmentation, researchers also actively followed the above rules and did not make any network parameter adjustments to obtain more objective results [39, 51]. The data downloaded from the official website contains CT slices of the abdomen and liver of 40 patients, of which 20 patients' slices are labeled, and the remaining 20 patients' slices are not provided with labels. We had to refer to common methods in the literature on the CHAOS dataset, such as [52, 53]. 1594 slices out of 20 patients are taken as the first dataset for our network. It is randomly divided into 1207, 257, and 130 samples, which are used for training, verification and testing, respectively.

**Lung** The lung data set was initiated by the National Cancer Institute (NCI) and jointly completed by the Lung

Imaging Database Consortium (LIDC) and the Imaging Database Resource Initiative (IDRI). We randomly select 192 out of 267 samples for training, 25 images as verification, and 50 as tests.

**ISIC 2018** The ISIC 2018 dataset is provided by the organizer of "MICCAI 2018 227 Challenge ISIC" for melanoma detection task. This article selects 2306 samples in the dataset for training, 200 samples for verification and testing.

## Evaluation index

In order to quantitatively evaluate the segmentation ability of MS-Net, the paper compares and evaluates six network models widely used in medical image segmentation. They are U-Net [27], U-Net + + [28] Attention-Unet [33], Channel-Unet [34], CE-Net [35] CS-Net [36] and VCA-Net [48]. To facilitate the intuitive comparison of these networks in these three datasets, the above networks have been retrained. Furthermore, five evaluation indicators are used for quantitative evaluation, including Dice similarity coefficient, (Miou) average intersection ratio, (ACC) accuracy, and area under the curve evaluation (AUC). The calculation formulas of Miou, Dice, ACC are as follows:

$$Miou = \frac{TP}{TP + FP + FN} \quad (9)$$

$$Dice = \frac{2TP}{2TP + FP + FN} \quad (10)$$

$$ACC = \frac{TN + TP}{TN + TP + FP + FN} \quad (11)$$

In the above formula, TN, TP, FN, and FP represent true negative, true positive, false negative and false positive, respectively. These indicator variables are used to evaluate calculation performance. Besides, AUC which represents the area under the ROC curve is also adopted as another quantitative indicator for evaluating segmentation performance.

## Experimental environment

The MS-Net network proposed in this article is implemented on the Tesla k40 with 12G under the framework of pytorch = 1.2.0. During the experiment, the Adam algorithm was selected as the optimization method, the batch size was set to 4, and all experiments were trained on the Ubuntu 18.04 operating system for 40 epochs.

**Table 1** Quantitatively compare the network model proposed in this article with six other advanced network models on the CHAOS dataset

| Method         | Miou (%)     | Dice (%)     | Auc (%)      | Acc (%)      |
|----------------|--------------|--------------|--------------|--------------|
| U-Net          | 84.99        | 91.74        | 96.07        | 98.09        |
| U-Net + +      | 85.23        | 91.76        | 96.96        | 98.15        |
| CS-Net         | 87.90        | 93.43        | 96.70        | 98.43        |
| CE-Net         | 88.54        | 93.74        | 96.65        | 98.59        |
| Attention-Unet | 88.37        | 93.67        | 96.81        | 98.58        |
| Channel-Unet   | 85.23        | 91.87        | 96.97        | 98.08        |
| VCA-Net        | 85.86        | 92.07        | <b>97.60</b> | 98.16        |
| MS-Net         | <b>89.87</b> | <b>94.54</b> | 97.09        | <b>98.69</b> |

## Results and analysis

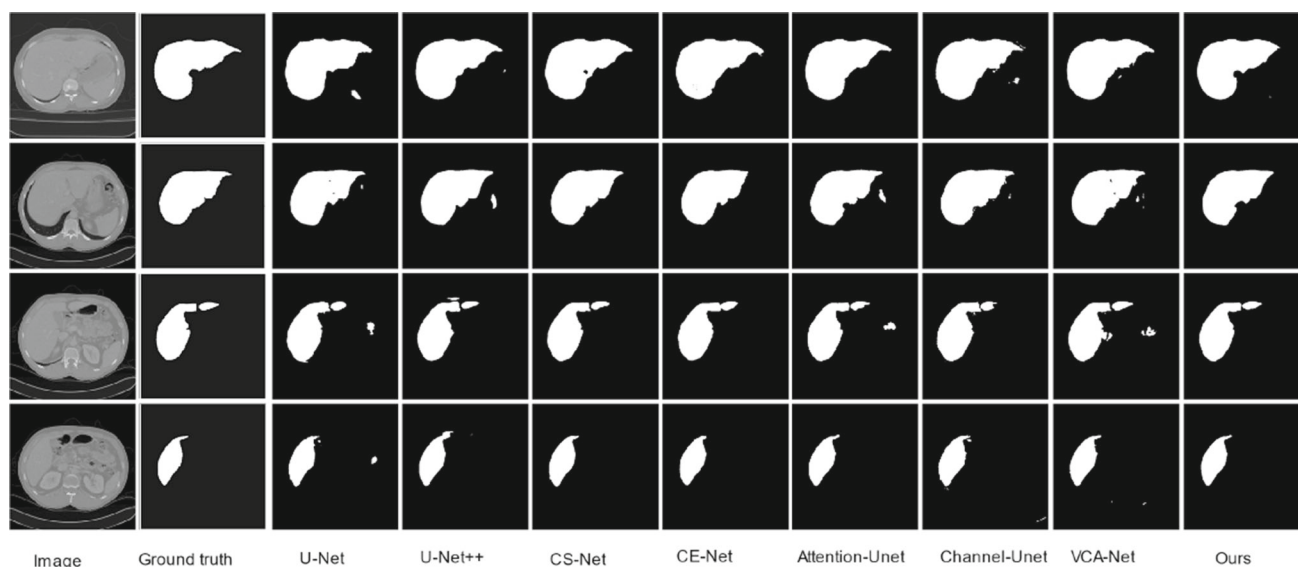
### Results on the CHAOS dataset

To demonstrate the segmentation performance of the proposed network, we first evaluate it on the CHAOS dataset and select six state-of-the-art methods for comparison, including: Two specially designed attention networks (Attention-Unet [33] and Channel-Unet [34]), two context codec networks (CE-Net [35], CS-Net [36] and VCA-Net [54]) and two most Classic networks (U-Net [27], U-Net + + [28]). For a fair comparison, the results of all our networks are learned in the same experimental environment.

Table 1 shows the segmentation results of different networks on the liver dataset. The data show that MS-Net scored the highest among the five indicators. Therefore, the results show that the method in this paper is effective in segmenting abdominal organs. As shown in Fig. 6, in order to visually compare MS-Net and other advanced methods, we take out four segmentation effect pictures from each model. As shown in Fig. 6, it can be observed that U-Net, U-Net + +, and other methods have severe segmentation. In contrast, the method in this paper can extract continuous and complete edge information. This shows that the network can obtain effective context information and can better deal with the problem of blurred edges. All in all, the performance of the method proposed in this paper is much higher than other state-of-the-art methods. Bold numbers in Table indicate highest scoring data.

### Results on the lung dataset

In order to verify the segmentation effect of our method on other morphological medical images, we also conducted comparative experiments on lung datasets. Comparison networks include: U-Net [27], U-Net + + [28], Attention-Unet [33], Channel-Unet [34], CE-Net [35], CS-Net [36] and VCA-Net [54]. The data in Table 2 show that the scores of the network proposed in this article are higher than other networks. As shown in Fig. 7, we observe that Channel-

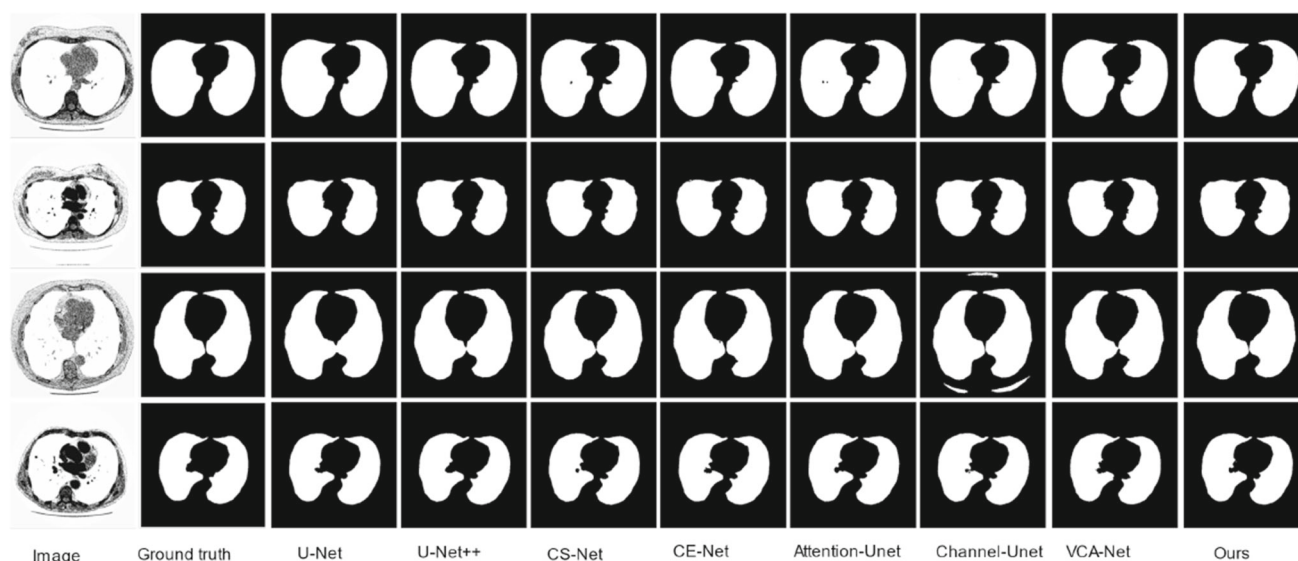


**Fig. 6** Some segmentation results obtained on the CHAOS dataset through the method in this article and other networks

Unet incorrectly increases the intensity of background pixels, resulting in over-segmentation. In contrast, the proposed MS-Net can adaptively learn the weights of pixels of interest and suppress some background features. However, the ability of the method in this paper to deal with some detailed features is not enough. From the segmentation effect demonstrated by the Row three and Col three in Fig. 7, it can be seen that the ability of the method in this paper is slightly lower than that of U-Net ++ in edge detail features.

**Table 2** Quantitatively compare the network model proposed in this article with the other six advanced network models on the Lung dataset

| Method         | Miou (%)     | Dice (%)     | Auc (%)      | Acc (%)      |
|----------------|--------------|--------------|--------------|--------------|
| U-Net          | 95.93        | 97.59        | 98.42        | 98.61        |
| U-Net ++       | 95.62        | 97.43        | 98.51        | 98.56        |
| CS-Net         | 95.19        | 97.21        | 98.42        | 98.48        |
| CE-Net         | 95.59        | 97.42        | 98.41        | 98.54        |
| Attention-Unet | 95.98        | 97.62        | 98.52        | 98.63        |
| Channel-Unet   | 95.14        | 97.19        | 98.59        | 98.43        |
| VCA-Net        | 95.79        | 97.53        | 98.69        | 98.58        |
| MS-Net (ours)  | <b>97.22</b> | <b>98.59</b> | <b>99.13</b> | <b>99.38</b> |



**Fig. 7** Some segmentation results obtained on the Lung dataset through the method of this article and other networks



**Table 3** Quantitatively compare the method proposed in this article with other six advanced network models on the ISIC 2018 dataset

| Method         | Miou (%)     | Dice (%)     | Auc (%)      | Acc (%)      |
|----------------|--------------|--------------|--------------|--------------|
| U-Net          | 73.69        | 82.55        | 89.39        | 87.96        |
| U-Net + +      | 74.99        | 83.68        | 89.82        | 88.98        |
| CS-Net         | 66.23        | 76.21        | 86.79        | 83.85        |
| CE-Net         | 77.55        | 85.94        | 91.17        | 90.96        |
| Attention-Unet | 71.53        | 81.03        | 88.61        | 86.27        |
| Channel-Unet   | 75.70        | 83.78        | 90.11        | 89.47        |
| VCA-Net        | 72.43        | 80.83        | 87.56        | 87.80        |
| MS-Net (ours)  | <b>78.00</b> | <b>86.06</b> | <b>91.24</b> | <b>91.29</b> |

### Results on the ISIC 2018 dataset

In order to verify the generalization ability of the network in this article, we will evaluate the method proposed in this article on another type of lesion data, also using U-Net [27], U-Net + + [28], Attention-Unet [33], Channel-Unet [34], CE-Net [35], CS-Net [36], VCA-Net [54] and MS-Net are compared on the skin lesion dataset. It is shown Table 3 that propose MS-Net achieves 86.06% on the Dice index, leading U-Net and CS-Net by 3.51% and 9.85%, respectively, which is much higher than other methods. The Dice index indicates the degree of overlap between the predicted result and the true label, indicating that the method in this paper can effectively obtain the complete lesion area compared to other methods. In the second column of Fig. 8, U-Net cannot accurately identify the lesion area when processing datasets such as skin lesions, and there is a serious under-segmentation phenomenon. Only the learning effect of the method in this article can achieve the best effect. Overall,

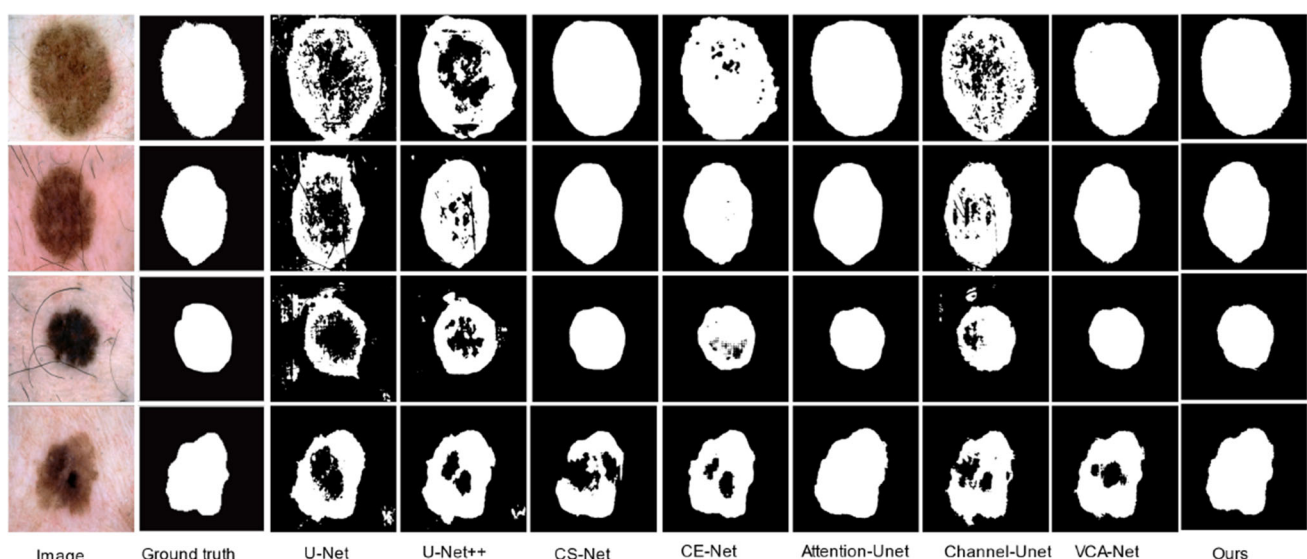
our MS-Net has obtained the best segmentation effect on the ISIC 2018 dataset, and the overall performance is ahead of other methods.

### Results on the ISIC 2018 dataset

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### Ablation experiment

This article conducts ablation experiments on MS-Net in the CHAOS dataset to evaluate the effectiveness of the MSAM and SFPM. The experimental data are shown in Table 4. First, when we added MSAM to the backbone network, Miou and

**Fig. 8** Some segmentation results obtained on the ISIC 2018 dataset through the method of this article and other networks

**Table 4** The ablation experiment of MS-Net on CHAOS data

| Backbone | MSAM | SFPM | Miou (%)     | Dice (%)     | Auc (%)      | Acc (%)      | parameter       |
|----------|------|------|--------------|--------------|--------------|--------------|-----------------|
| ✓        |      |      | 84.99        | 91.74        | 96.07        | 98.09        | 7.766 M         |
| ✓        | ✓    |      | 87.32        | 93.08        | 96.85        | 98.34        | 11.069 M        |
| ✓        |      | ✓    | 87.94        | 93.44        | 96.31        | 98.49        | 19.182 M        |
| ✓        | ✓    | ✓    | <b>89.87</b> | <b>94.54</b> | <b>97.09</b> | <b>98.69</b> | <b>22.486 M</b> |

**Table 5** Ablation experiment of FPM on CHAOS dataset

| Network          | Miou (%)     | Dice (%)     | Auc (%)      | Acc (%)      |
|------------------|--------------|--------------|--------------|--------------|
| Ours without FPM | 86.00        | 92.29        | 96.19        | 98.27        |
| Ours             | <b>89.87</b> | <b>94.54</b> | <b>97.09</b> | <b>98.69</b> |

**Table 6** Comparison of different parameter strategies on MSAM

| Model | Parameter |           |           | Miou (%)     | Dice (%)     | Auc (%)      | Acc (%)      |
|-------|-----------|-----------|-----------|--------------|--------------|--------------|--------------|
|       | 2th layer | 3th layer | 4th layer |              |              |              |              |
| MSAM  | (1,3)     | (1,5)     | (1,7)     | 86.97        | 92.66        | 96.01        | 98.36        |
|       | (1,3)     | (1,5)     | (3,5)     | 89.27        | 94.10        | 96.15        | 98.64        |
|       | (1,3,5)   | (1,3,7)   | (1,5,7)   | 87.11        | 92.94        | 96.29        | 98.34        |
|       | (1,3,5)   | (1,3,7)   | (3,5,7)   | 86.07        | 92.29        | 96.52        | 98.22        |
|       | (1,3)     | (3,5)     | (5,7)     | <b>89.87</b> | <b>94.54</b> | <b>97.09</b> | <b>98.69</b> |

Dice increased by 2.33% and 1.34%, respectively. This shows that the introduction of MSAM has improved the network's ability to collect features at different levels. Secondly, we also embed SFPM into the backbone network for experiments. Miou and Dice rose 2.95% and 1.7%, respectively. This shows that SFPM can enhance the sensitivity of the network to the features of interest, so that the network can effectively capture the features. However, according to the data in Table 4, it can be seen that SFPM sacrifices a lot of parameters in order to improve the front-end effect of the network, which often consumes a lot of time and memory. This module deserves further exploration in future research.

In addition, we also conducted another ablation experiment to evaluate the contribution of the FPM to the network model of this paper. The data in Table 5 show that after removing FPM, the Dice index dropped by 1.68%, and other indicators weakened. This shows that the sub-module FPM has a very important auxiliary role in the SFPM. It can effectively extract the characteristics of the lesion area, thereby further improving the performance of the network.

Finally, we conduct parameter comparison experiments on MSAM and SFPM modules. As shown in Tables 6 and 7, the two channels with different dilated rate in MSAM can already improve the network accuracy. Further increases of the channel number cannot deliver significant performance improvements, but does increase the parameters. Therefore,

**Table 7** Comparison of different parameter strategies on SFPM

| Model | Parameter |          |          | Miou (%)     | Dice (%)     | Auc (%)      | Acc (%)      |
|-------|-----------|----------|----------|--------------|--------------|--------------|--------------|
| SFPM  | 1         | 3        | 5        | 88.52        | 93.69        | <b>97.70</b> | 98.62        |
|       | <b>1</b>  | <b>2</b> | <b>4</b> | <b>89.87</b> | <b>94.54</b> | 97.09        | <b>98.69</b> |

in the skip connection data of layers 2, 3, and 4, the parameters are selected as (1, 3), (3, 5) and (5, 7), respectively. For SFPM, when the parallelism parameters are set to 1, 2 and 4, the performances of Miou, Dice and Acc are higher, except for the Auc is slightly lower. In addition, we provide a glossary of all acronyms shown in Table 8.

## Conclusion and future works

A new MS-Net framework based on the codec structure is proposed in this paper, which is used for the segmentation task of medical lesions. This article mainly embeds two core modules in the backbone network, namely MSAM and SFPM. Firstly, MSAM was developed as skip connections to extract different levels of contextual details. Secondly, the SFPM is nested in the deepest part of the network, which can process features of different scales at the same time, allowing the network to better focus on the target area. Finally, by applying a newly introduced sub-module of SFPM named FPM to adaptively increase the weight of the features of interest, the segmentation effect has been significantly improved. Three datasets (CHAOS, Lung and ISIC 2018) were used to test the proposed MS-Net, and scores of 94.54%, 98.59% and 86.06% have been achieved in Dice scores, leading U-Net by 2.8%, 1% and 3.51%, respectively. Additionally, the comparison with six other most advanced networks also shows that the proposed network has great advantages in evaluation indicators such as Miou, Dice, Acc and Auc. The experimental results fully demonstrate that the designed network can achieve better results in most medical segmentation tasks.

However, since 3D structures are expected to achieve higher performance or are even mandatory in some segmentation scenarios, future attempts could be made to transfer our structure to high dimensional tasks. To this end, further efforts are needed on how to replace 2D convolution operations with 3D ones, simplify the network architecture and reduce the network parameters.

**Table 8** Nomenclature

|                |                                                                                     |                      |                                                                         |
|----------------|-------------------------------------------------------------------------------------|----------------------|-------------------------------------------------------------------------|
| Attention-Unet | Learning where to look for the pancreas                                             | <i>NCI</i>           | The National Cancer Institute                                           |
| CE-Net         | Context encoder network for 2d medical image segmentation                           | <i>STAPLE</i>        | Simultaneous Truth and Performance Level Estimation                     |
| Channel-Unet   | spatial channel-wise convolutional neural network for liver and tumors segmentation | <i>SFPM</i>          | Stacked Feature Pyramid Module                                          |
| CS-Net         | channel and spatial attention network for curvilinear structure segmentation        | <i>U-Net</i>         | Convolutional Networks for Biomedical Image Segmentation                |
| CT             | Computed Tomography                                                                 | <i>U-Net + +</i>     | A nested u-net architecture for medical image segmentation              |
| FPM            | Feature Perception Module                                                           | <i>VCA-Net</i>       | Stroke Lesion Segmentation with Visual Cortex Anatomy Alike Neural Nets |
| TN             | True Negative                                                                       | <i>C</i>             | Concatenation                                                           |
| TP             | True Positive                                                                       | <i>F<sub>i</sub></i> | Feature map                                                             |
| FN             | False Negative                                                                      | <i>d<sub>i</sub></i> | Dilated convolution                                                     |
| FP             | False Positive                                                                      | <i>f<sub>i</sub></i> | Weights assigned to feature maps after learning                         |
| IDRI           | the Imaging Database Resource Initiative                                            | <i>S</i>             | Softmax                                                                 |
| LIDC           | the Lung Imaging Database Consortium                                                | $\oplus$             | Element addition                                                        |
| MSAM           | Multi-Scale Attention Module                                                        | $\otimes$            | Multiplication                                                          |
| MS-Net         | Multi-Scale Feature Pyramid Fusion Network                                          | $\odot$              | matrix multiplication                                                   |
| MV             | Majority Voting                                                                     |                      |                                                                         |

**Acknowledgements** The work was supported in part by the National Natural Science Foundation of China with Grant 62065003. Supported by Guizhou Provincial Science and Technology Projects -ZK [2022] Key-020.

**Conflict of interest** The authors declare that they have no conflict of interest.

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